



Habib University

School of Science & Engineering

Course:	EE – 211 – Basic Electronics
Semester	Fall 2024
Assignment #	2
Due Date :	April 8, 2024
Instructor :	Syed Arsalan Jawed
Total Marks :	100

Instructions for Students

1. Print the following question paper and provide your answer within the empty spaces given after each question.
2. Keep your answers concise and to the point. Provide neat and tidy calculations and plots.
3. Make reasonable assumptions related to concepts and associated parameters motivated by your text and reference book.
4. Support your answer with equations where needed.

Question # 1 : [CLO-1]

Consider an NPN BJT. The junction forward bias voltage is 0.8V. You have two configurable supply voltages, which can be configured to any value between 0V and 5V. Make the BJT operate in active mode first by selecting appropriate V_{BE} and V_{CB} . Draw the NPN structure and show the connected voltage sources. Then similarly, make the BJT operate in saturation mode and share the structure and connection. Show the current flows inside the NPN structure in both cases. [20]

Soln

Active Mode

Select $V_{BE}=0.8V$ to forward bias the Base Emitter Junction

Select $V_{CB} > 0$ to reverse bias the collector base junction. V_{CB} should be such that $V_{CE}(\text{sat}) > 0.4V$ to stay away from edge of saturation region, we may chose $V_{CE} = 2V$ as we have supply till 5V.

$$V_{CE}(\text{sat}) > 0.4V, V_{CE} = 2V$$

$$V_{BE} = 0.8V$$

$$V_{CB} > V_{CE} - V_{BE} = 2 - 0.8 > 1.2V$$

Saturation Mode

For deep inside the saturation region, $V_{CE}(\text{sat}) = 0.2V$

$$V_{BE} = 0.8V$$

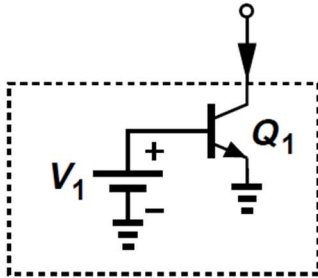
$$V_{CB} = V_{CE} - V_{BE} = 0.2 - 0.8 = -0.6V$$

$$V_{BC} = 0.6V \text{ (base collector junction forward bias by } 0.6V)$$

Refer to your book for structure and current flow.

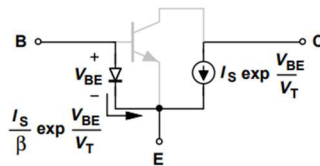
Question # 2 : [CLO-1]

Draw the large-signal model of the following BJT. Then convert the large-signal model into small-signal model. Ignore Early effect and derive the expression for transconductance. [20]



Large Signal Model (refer to the mentioned eqns and figures in your text book)

Since the base-emitter junction is forward-biased in the active mode, we can place a diode between the base and emitter terminals. Moreover, since the current drawn from the collector and flowing into the emitter depends on only the base-emitter voltage, we add a voltage-controlled current source between the collector and the emitter, arriving at the model shown in Fig. 4.13. As illustrated in Fig. 4.11, this current remains independent of the collector-emitter voltage. But how do we ensure that the current flowing through the diode is equal to $1/\beta$ times the collector current? Equation (4.24) suggests that the base current is equal to that of a diode having a reverse saturation current of I_S/β . Thus, the base-emitter junction is modeled by a diode whose cross section area is $1/\beta$ times that of the actual emitter area.



The derivation of the small-signal model from the large-signal counterpart is relatively straightforward. We perturb the voltage difference between every two terminals (while the third terminal remains at a constant potential), determine the changes in the currents flowing through *all* terminals, and represent the results by proper circuit elements such as controlled current sources or resistors. Figure 4.21 depicts two conceptual examples where V_{BE} or V_{CE} is changed by ΔV and the changes in I_C , I_B , and I_E are examined.

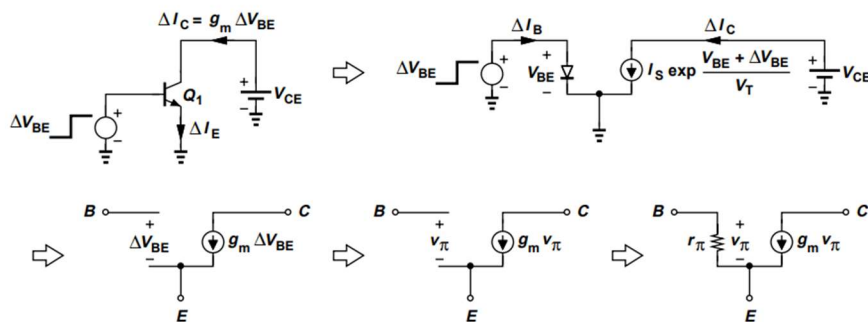
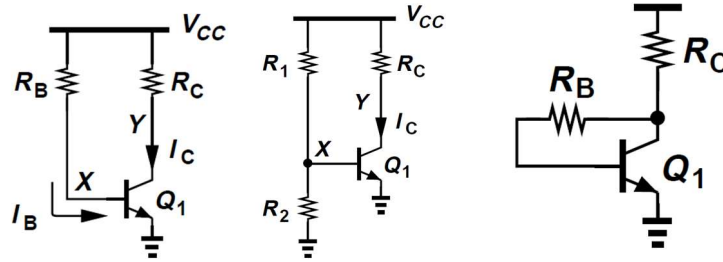


Figure 4.22 Development of small-signal model.

Question # 3 : [CLO2]

Q3. Derive the expression for the bias voltage at the base of the following BJT bias circuits. Compare each approach by listing one advantage and one disadvantage of each approach. [20]



(a) $V_B = V_{CC} - I_B R_B = V_{CC} - I_C R_B / \beta = V_{CC} - \{(V_{CC} - V_{CE}) / R_C\} R_B / \beta$
 $V_{CE} \geq 0.4V$

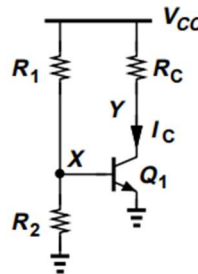
The biasing scheme of Fig. 5.13 merits a few remarks. First, the effect of V_{BE} “uncertainty” becomes more pronounced at low values of V_{CC} because $V_{CC} - V_{BE}$ determines the base current. Thus, in low-voltage design—an increasingly common paradigm in modern electronic systems—the bias is more sensitive to V_{BE} variations among transistors or with temperature. Second, we recognize from Eq. (5.15) that I_C heavily depends on β , a parameter that may change considerably. In the above example, if β increases from 100 to 120, then I_C rises to 1.98 mA and V_{CE} falls to 0.52, driving the transistor toward heavy saturation. For these reasons, the topology of Fig. 5.13 is rarely used in practice.

$$V_X = \frac{R_2}{R_1 + R_2} V_{CC}, \quad (5.29)$$

if the base current is negligible. Thus,

$$I_C = I_S \exp \left(\frac{R_2}{R_1 + R_2} \cdot \frac{V_{CC}}{V_T} \right), \quad (5.30)$$

a quantity independent of β . Nonetheless, the design must ensure that the base current remains negligible.



(b)

We now determine the collector bias current by assuming $I_B \ll I_C$; i.e., R_C carries a current equal to I_C , thereby yielding

$$V_Y = V_{CC} - R_C I_C. \quad (5.75)$$

Also,

$$V_Y = R_B I_B + V_{BE} \quad (5.76)$$

$$= \frac{R_B I_C}{\beta} + V_{BE}. \quad (5.77)$$

Equating the right hand sides of (5.75) and (5.77) gives

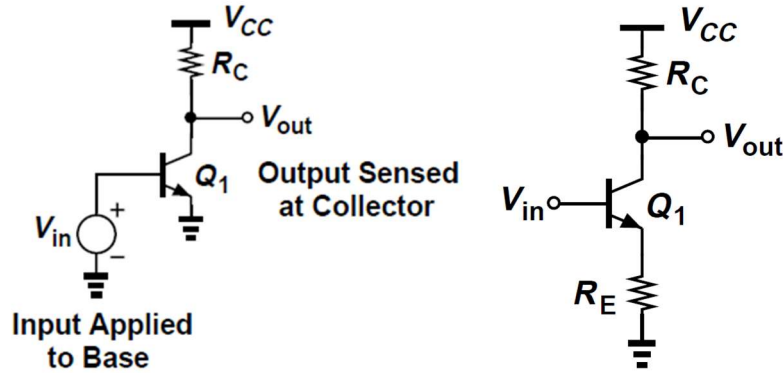
$$I_C = \frac{V_{CC} - V_{BE}}{R_C + \frac{R_B}{\beta}}. \quad (5.78)$$

(c)

Always in active mode.

Question # 4 : [CLO-2]

Derive the expressions for small-signal voltage gain, input impedance and output impedance for the following common emitter configurations. [20]



Let us first compute the small-signal voltage gain $A_v = v_{out}/v_{in}$. Beginning from the output port and writing a KCL at the collector node, we have

$$-\frac{v_{out}}{R_C} = g_m v_\pi, \quad (5.113)$$

and $v_\pi = v_{in}$. It follows that

$$(a) \quad A_v = -g_m R_C. \quad (5.114)$$

$$g_m v_\pi = -\frac{v_{out}}{R_C}, \quad (5.149)$$

obtaining

$$v_\pi = -\frac{v_{out}}{g_m R_C}. \quad (5.150)$$

We also recognize that two currents flow through R_E : one originating from r_π equal to v_π/r_π and another equal to $g_m v_\pi$. Thus, the voltage drop across R_E is given by

$$v_{RE} = \left(\frac{v_\pi}{r_\pi} + g_m v_\pi \right) R_E. \quad (5.151)$$

Since the voltage drop across r_π and R_E must add up to v_{in} , we have

$$v_{in} = v_\pi + v_{RE} \quad (5.152)$$

$$= v_\pi + \left(\frac{v_\pi}{r_\pi} + g_m v_\pi \right) R_E \quad (5.153)$$

(b)

$$= v_{\pi} \left[1 + \left(\frac{1}{r_{\pi}} + g_m \right) R_E \right]. \quad (5.154)$$

Substituting for v_{π} from (5.150) and rearranging the terms, we arrive at

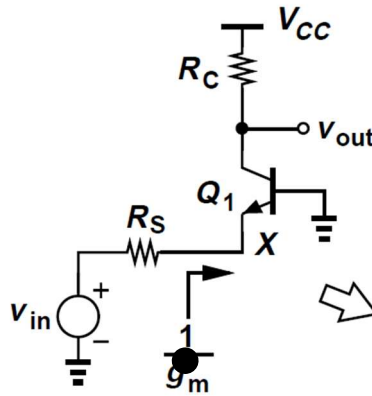
$$\frac{v_{out}}{v_{in}} = - \frac{g_m R_C}{1 + \left(\frac{1}{r_{\pi}} + g_m \right) R_E}. \quad (5.155)$$

As predicted earlier, the magnitude of the voltage gain is lower than $g_m R_C$ for $R_E \neq 0$. With $\beta \gg 1$, we can assume $g_m \gg 1/r_{\pi}$ and hence

$$A_v = - \frac{g_m R_C}{1 + g_m R_E}. \quad (5.156)$$

Question # 5 : [CLO-3]

Consider the following circuit where a source V_{in} is providing input to the common-base amplifier topology core (biasing is not shown) where R_s is the output impedance of the driving source. We need to input impedance of the amplifier with R_s . $R_s=50\Omega$. Suggest which small-signal parameter of the amplifier should be aligned to achieve matching input impedance, suggest what large-signal parameters that define the operating point should be used to achieve the required small signal parameter, give exact value of each parameter with supporting expressions. [20]



Let us now compute the I/O impedances of the CB topology so as to understand its capabilities in interfacing with preceding and following stages. The rules illustrated in Fig. 5.7 prove extremely useful here, obviating the need for small-signal equivalent circuits. Shown in Fig. 5.64(a), the simplified ac circuit reveals that R_{in} is simply the impedance seen looking into the emitter with the base at ac ground. From the rules in Fig. 5.7, we have

$$R_{in} = \frac{1}{g_m} \quad (5.263)$$